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Scenario: AWS VPC Peering with Two Instances in Private IP

Solution:

1. Go to the AWS Management Console, search for VPC, and Create a new one VPC as follows

The screenshot displays the AWS Management Console interface for creating a new VPC. The breadcrumb navigation shows 'VPC > Your VPCs > Create VPC'. The main heading is 'Create VPC' with an 'Info' link. A descriptive text states: 'A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.'

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

vpc-01

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.1.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q vpc-01 X Remove tag

Add tag

You can add 49 more tags

Cancel Create VPC

3. Create another VPC as shown below.

Services

Search

[Alt+S]

VPC

>

Your VPCs

>

Create VPC

Create VPC

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only

☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

vpc-02

IPv4 CIDR block

☒ IPv4 CIDR manual input

☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.2.0.0/14

CIDR block size must be between /16 and /28.

IPv6 CIDR block

☒ No IPv6 CIDR block

☐ IPAM-allocated IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Tenancy

Default

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name

X

Q vpc-02

X

Remove tag

Add tag

You can add 49 more tags

Cancel

Create VPC

CloudShell

Feedback

4. Create Subnet for VPC-2 as shown below.

aws

Services

Search

[Alt+S]

VPC > Subnets > Create subnet

Create subnet [Info](#)

VPC

VPC ID

Create subnets in this VPC.

vpc-0a5b657387cde4f02 (vpc-02)

Associated VPC CIDRs

IPv4 CIDRs

10.2.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

vpc-02

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

US East (N. Virginia) / us-east-1a

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.2.0.0/16

IPv4 subnet CIDR block

10.2.1.0/24 256 IPs

< > ^ v

▼ Tags - optional

Key

Value - optional

Q Name X

Q vpc-02 X

Remove

Add new tag

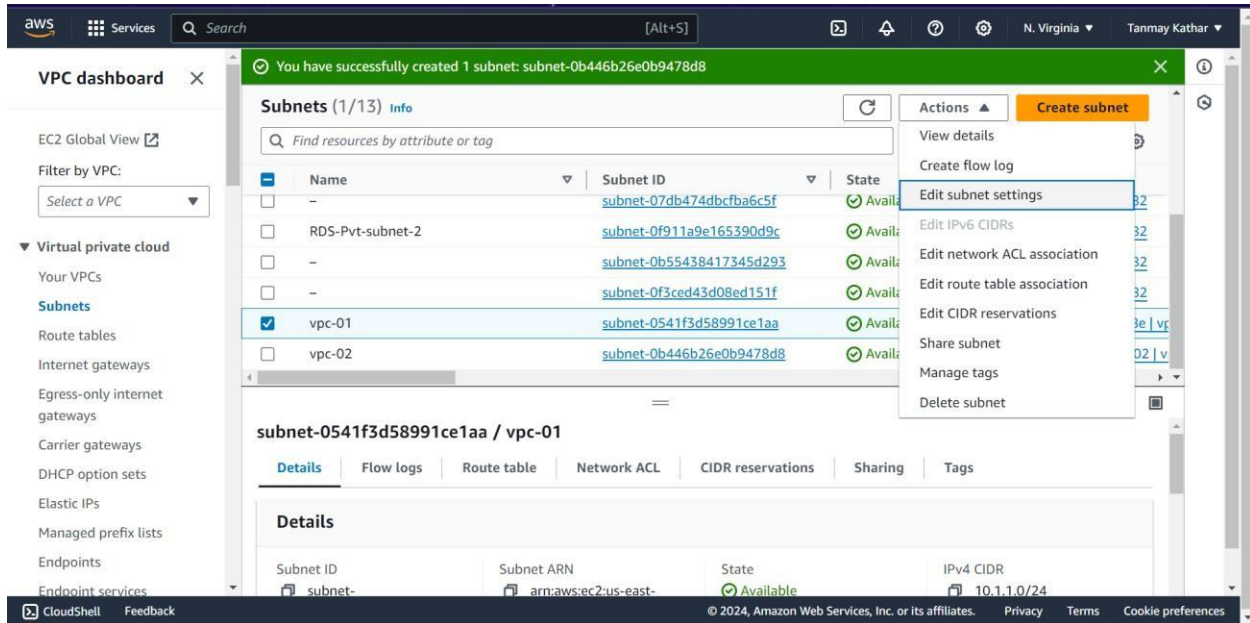
You can add 49 more tags.

Remove

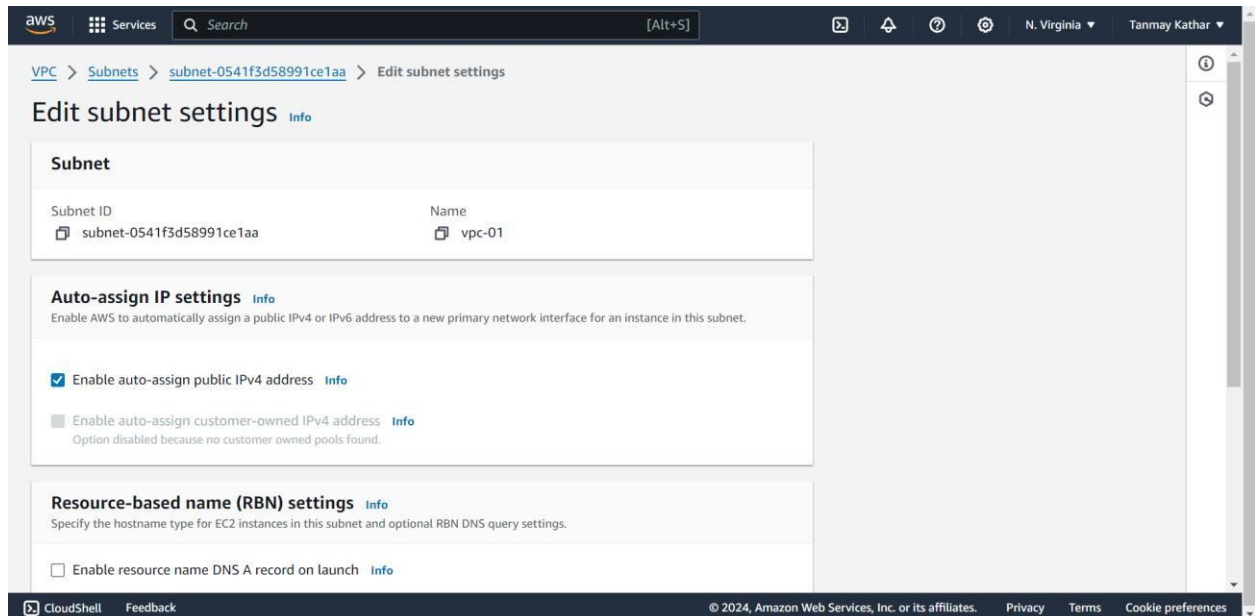
Add new subnet

CloudShell Feedback

5. Below are the VPCs that we have created.



6. Now edit both Subnets we have already created and “enable auto-assign public IPv4 address” .



aws

Services

Search

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VPC > Subnets > subnet-0b446b26e0b9478d8 > Edit subnet settings

Edit subnet settings [Info](#)

Subnet

Subnet ID

Subnet-0b446b26e0b9478d8

Name

vpc-02

Auto-assign IP settings [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

☐ Enable auto-assign customer-owned IPv4 address [Info](#)
Option disabled because no customer owned pools found.

Resource-based name (RBN) settings [Info](#)

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch [Info](#)

CloudShell Feedback

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7. Create Internet Gateway

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Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

custom-internet-gw

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X

Q custom-internet-gw X

Remove

Add new tag

You can add 49 more tags.

Cancel **Create internet gateway**

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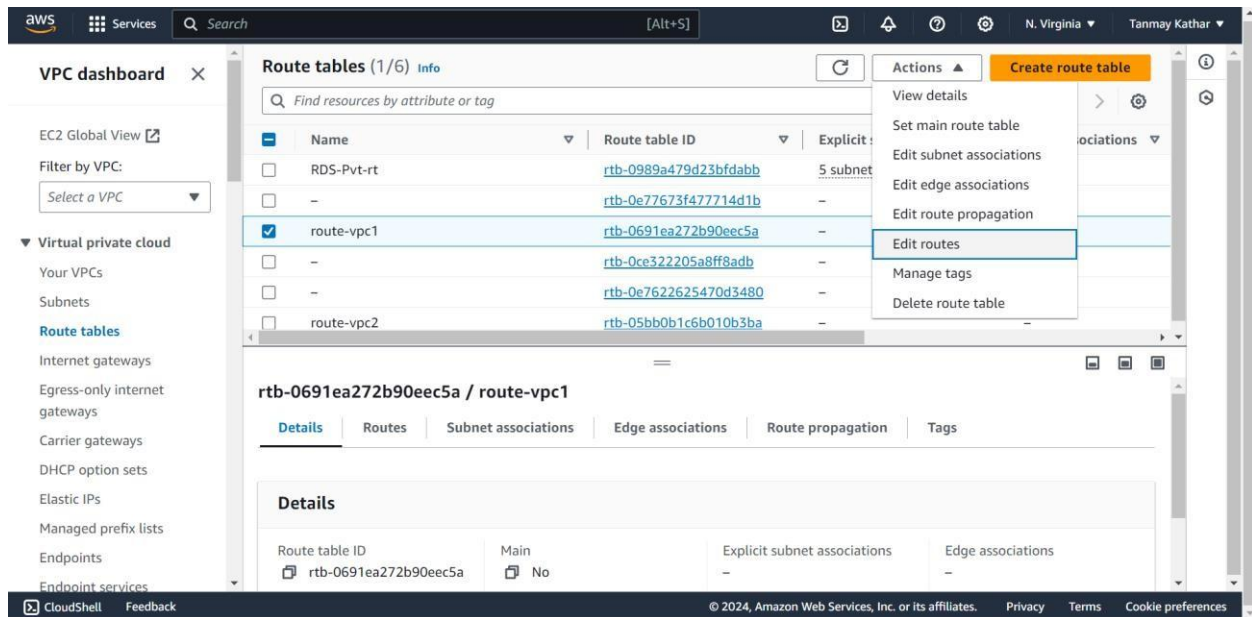
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8. Attach VPC to Internet Gateway.

The screenshot shows the AWS Management Console interface for the 'Attach to VPC' page. The breadcrumb navigation at the top reads 'VPC > Internet gateways > Attach to VPC (igw-072d8591a459b3919)'. The main heading is 'Attach to VPC (igw-072d8591a459b3919)' with an 'Info' link. Below this, a section titled 'VPC' contains the instruction: 'Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.' Underneath, the 'Available VPCs' section prompts the user to 'Attach the internet gateway to this VPC.' and features a search input field containing 'vpc-09fcc63c365114a8d'. A section for the 'AWS Command Line Interface command' is partially visible. At the bottom of the main content area are two buttons: 'Cancel' and 'Attach internet gateway'.

9. Create Route table for both VPCs.

The screenshot displays the AWS Management Console 'Create route table' page. The breadcrumb navigation is 'VPC > Route tables > Create route table'. The main heading is 'Create route table' with an 'Info' link. A descriptive paragraph states: 'A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.' The 'Route table settings' section includes a 'Name - optional' field with the value 'route-vpc2' and a 'VPC' dropdown menu set to 'vpc-0a5b657387cde4f02 (vpc-02)'. The 'Tags' section shows a list with one tag: 'Name' as the key and 'route-vpc2' as the value. At the bottom of the page, there is an 'Add another tag' button.



10. Create Peering Connection.

aws Services Search [Alt+S]

VPC > Peering connections > Create peering connection

Create peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately.

Info

Peering connection settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Select a local VPC to peer with

VPC ID (Requester)

VPC CIDRs for vpc-09fcc63c365114a8e (vpc-01)

CIDR	Status	Status reason
10.1.0.0/16	Associated	-

Select another VPC to peer with

Account
☒ My account
☐ Another account

Region
☒ This Region (us-east-1)
☐ Another Region

VPC ID (Acceptor)

VPC CIDRs for vpc-0a5b657387cde4f02 (vpc-02)

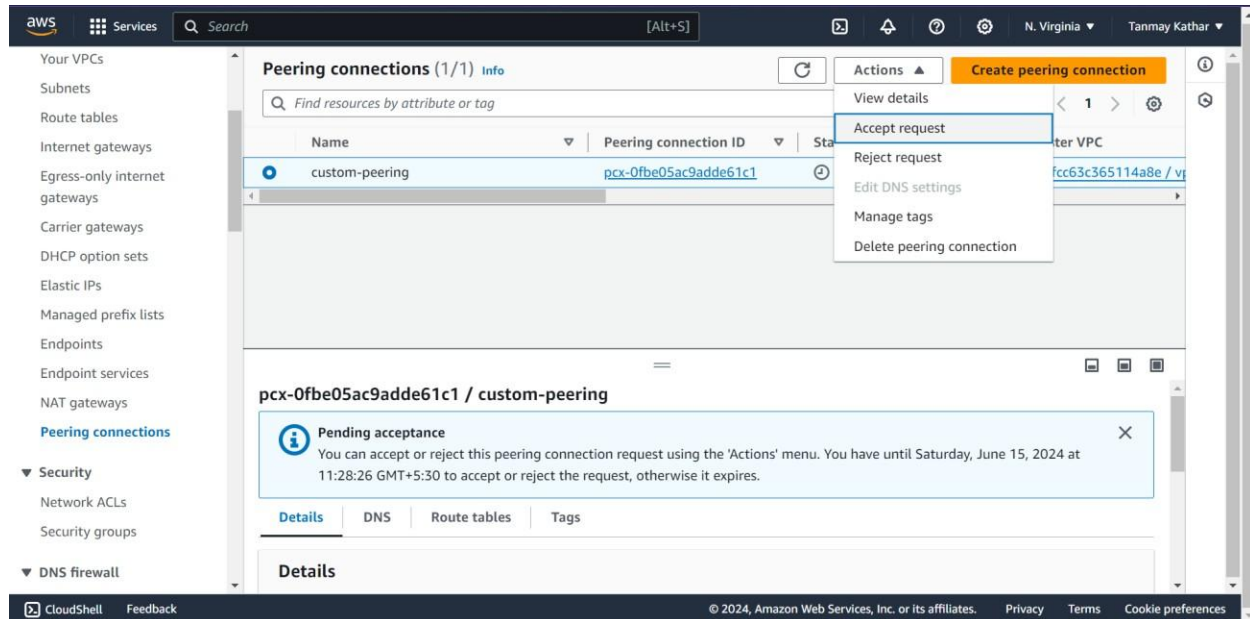
CIDR	Status	Status reason
10.2.0.0/16	Associated	-

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

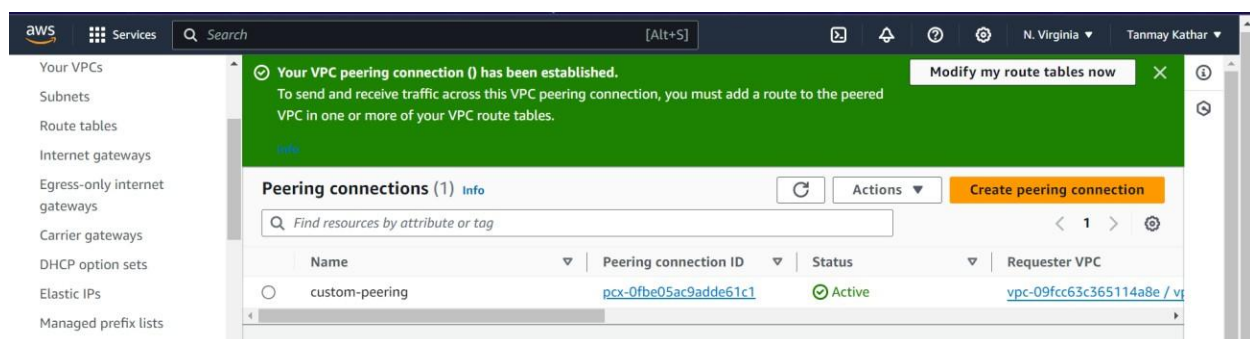
Key: Value - optional:

You can add 49 more tags.

11. Once VPC peering connection is creating go to “Action” and “Accept request” and you will see peering connection is “active”.



12. Click on Right corner “Modify my route table now” Edit Both Route table.



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VPC > Route tables > rtb-05bb0b1c6b010b3ba > Edit routes

Edit routes

Destination	Target	Status	Propagated	
10.2.0.0/16	local	Active	No	
0.0.0.0/0	local			
	Internet Gateway	Active	No	Remove
	igw-074942cbcd5c39705			
10.1.0.0/16	Peering Connection	-	No	Remove
	pcx-0fbc05ac9adde61c1			
Add route				

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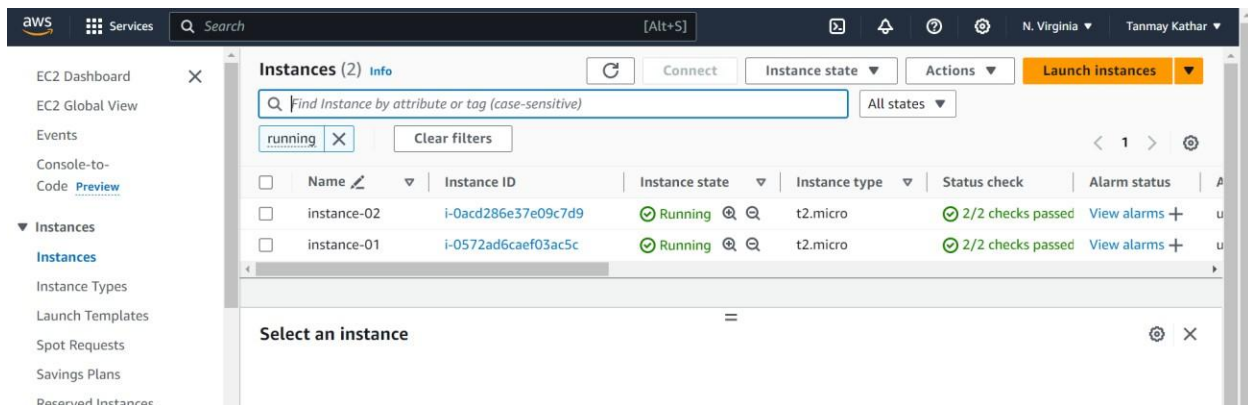
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VPC > Route tables > rtb-0691ea272b90eec5a > Edit routes

Edit routes

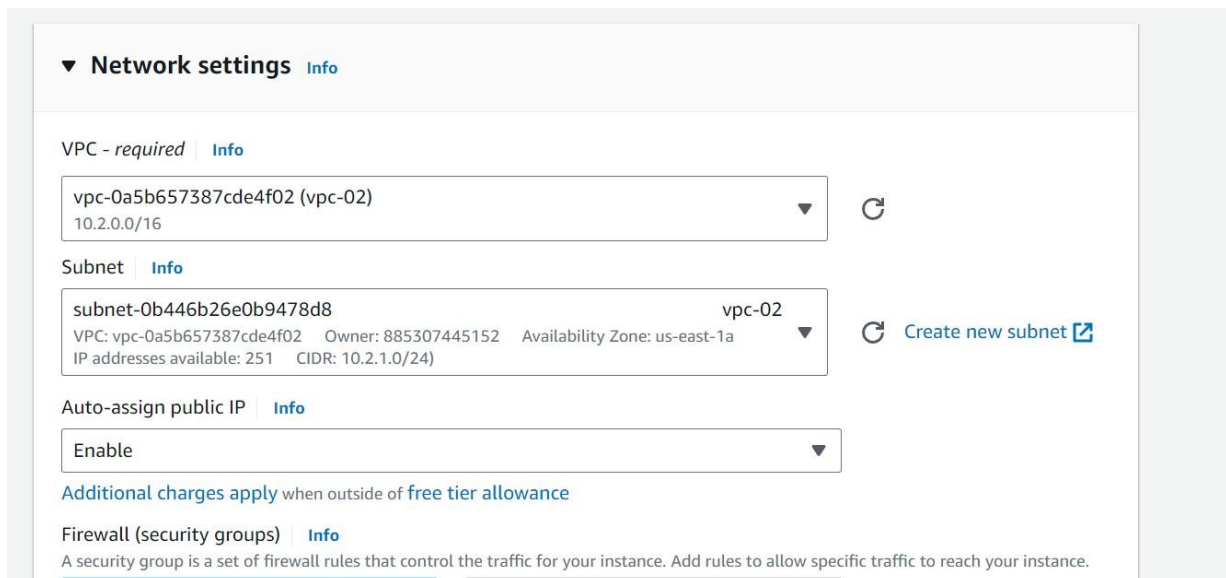
Destination	Target	Status	Propagated	
10.1.0.0/16	local	Active	No	
0.0.0.0/0	local			
	Internet Gateway	Active	No	Remove
	igw-072d8591a459b3919			
10.2.0.0/16	Peering Connection	-	No	Remove
	pcx-0fbc05ac9adde61c1			
Add route				

13. Create two instances to test VPC peering connection.



14. While Creating instance select VPCs that we have created already

- VPC-01= instance1
- VPC-02= instance2



15. Now Connect both instances and try to ping both instance using there IPs.

- Instance 1-IP: 10.1.1.193
- Instance 2-IP: 10.2.1.238

Install apache in “instance-01” and edit index.html file.

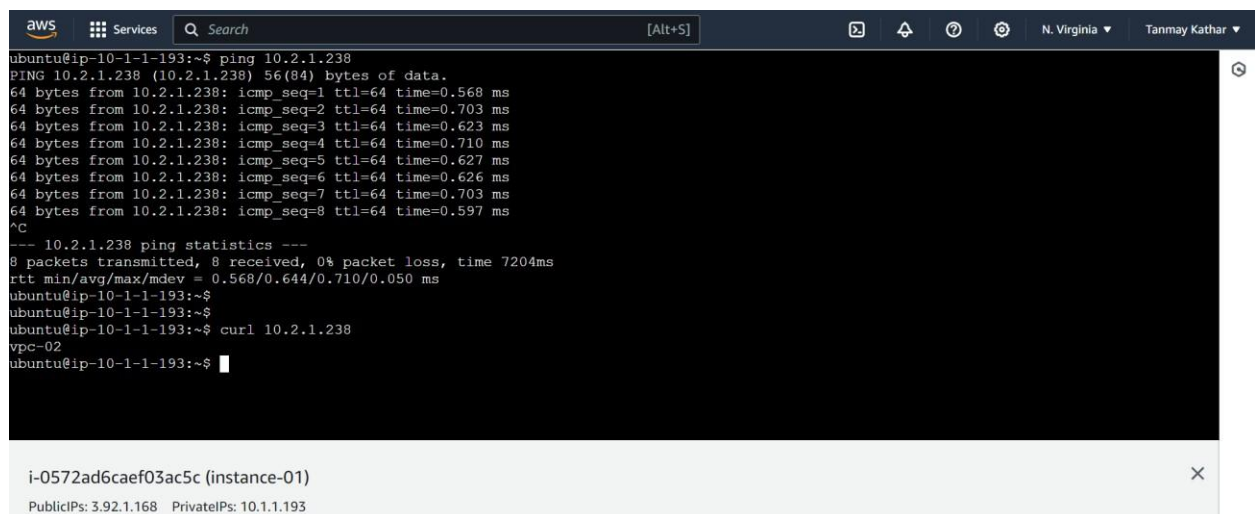
- `sudo apt install apache2 -y`
- `cd /var/www/html`
- `sudo echo “VPC-02” > index.html`
- `sudo systemctl enable apache2`
- `sudo systemctl restart apache2`

Install apache in “instance-02” and edit index.html file.

- `sudo apt install apache2 -y`
- `cd /var/www/html`
- `sudo echo “VPC-01” > index.html`
- `sudo systemctl enable apache2`
- `sudo systemctl restart apache2`

In instance-01

- `ping 10.2.1.238`
- `curl 10.2.1.238`



```
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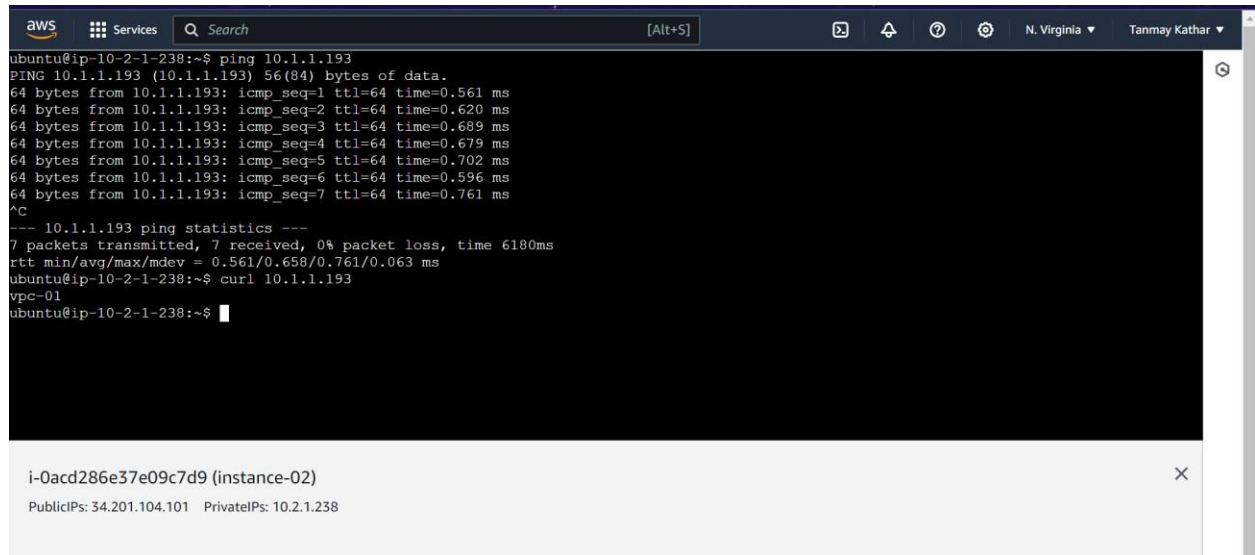
ubuntu@ip-10-1-1-193:~$ ping 10.2.1.238
PING 10.2.1.238 (10.2.1.238) 56(84) bytes of data:
64 bytes from 10.2.1.238: icmp_seq=1 ttl=64 time=0.568 ms
64 bytes from 10.2.1.238: icmp_seq=2 ttl=64 time=0.703 ms
64 bytes from 10.2.1.238: icmp_seq=3 ttl=64 time=0.623 ms
64 bytes from 10.2.1.238: icmp_seq=4 ttl=64 time=0.710 ms
64 bytes from 10.2.1.238: icmp_seq=5 ttl=64 time=0.627 ms
64 bytes from 10.2.1.238: icmp_seq=6 ttl=64 time=0.626 ms
64 bytes from 10.2.1.238: icmp_seq=7 ttl=64 time=0.703 ms
64 bytes from 10.2.1.238: icmp_seq=8 ttl=64 time=0.597 ms
^C
--- 10.2.1.238 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7204ms
rtt min/avg/max/mdev = 0.568/0.644/0.710/0.050 ms
ubuntu@ip-10-1-1-193:~$
ubuntu@ip-10-1-1-193:~$
ubuntu@ip-10-1-1-193:~$ curl 10.2.1.238
vpc-02
ubuntu@ip-10-1-1-193:~$
```

i-0572ad6caef03ac5c (instance-01)

PublicIPs: 3.92.1.168 PrivateIPs: 10.1.1.193

In instance-02

- **ping 10.1.1.193**
- **curl 10.1.1.193**



The screenshot shows an AWS Management Console terminal window for an Ubuntu instance. The terminal output shows a successful ping to 10.1.1.193 and a successful curl command. The terminal window has a dark background with white text. The AWS console header is visible at the top, showing the AWS logo, 'Services' menu, a search bar, and the user's name 'Tanmay Kathar'.

```
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ubuntu@ip-10-2-1-238:~$ ping 10.1.1.193
PING 10.1.1.193 (10.1.1.193) 56(84) bytes of data:
64 bytes from 10.1.1.193: icmp_seq=1 ttl=64 time=0.561 ms
64 bytes from 10.1.1.193: icmp_seq=2 ttl=64 time=0.620 ms
64 bytes from 10.1.1.193: icmp_seq=3 ttl=64 time=0.689 ms
64 bytes from 10.1.1.193: icmp_seq=4 ttl=64 time=0.679 ms
64 bytes from 10.1.1.193: icmp_seq=5 ttl=64 time=0.702 ms
64 bytes from 10.1.1.193: icmp_seq=6 ttl=64 time=0.596 ms
64 bytes from 10.1.1.193: icmp_seq=7 ttl=64 time=0.761 ms
^C
--- 10.1.1.193 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6180ms
rtt min/avg/max/mdev = 0.561/0.658/0.761/0.063 ms
ubuntu@ip-10-2-1-238:~$ curl 10.1.1.193
vpc-01
ubuntu@ip-10-2-1-238:~$
```

i-0acd286e37e09c7d9 (instance-02) X

PublicIPs: 34.201.104.101 PrivateIPs: 10.2.1.238